

Minimum Latency Training Strategies for Streaming Sequence-to-Sequence ASR

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IEEE ICASSP 2020



Background: End-to-end ASR

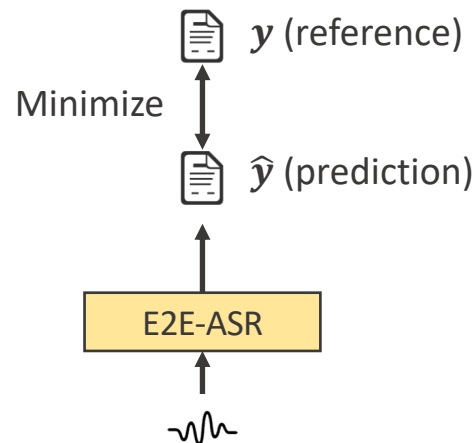
- Input sequence (speech): $\mathbf{x} = (x_1, \dots, x_T)$
- Output sequence (transcription): $\mathbf{y} = (y_1, \dots, y_L)$

Time-synchronous model ($|\mathbf{x}| = |\hat{\mathbf{y}}|$)

- Connectionist temporal classification (CTC) [Graves et al., 2006]
- RNN-Transducer (RNN-T) [Graves et al., 2013]
- Recurrent neural aligner (RNA) [Sak et al., 2017]

Label-synchronous model ($|\mathbf{x}| \neq |\hat{\mathbf{y}}|$)

- Attention-based sequence-to-sequence (S2S) [Bahdanau et al., 2016]
- Transformer [Vaswani et al., 2017]



Low accuracy
Streaming: easy

High accuracy
Streaming: difficult

Streaming attention-based S2S ASR

Neural Transducer [Jailty et al., 2015]

- Perform attention mechanism for a fixed size of block

Hard monotonic attention [Raffel et al., 2017]

- Learn to detect token boundaries via stochastic binary decision
- Extension: **Monotonic chunkwise attention (MoChA)** [Chiu et al., 2018]

Triggered attention [Moritz et al., 2018]

- Perform global attention over encoder memories trun



Adaptive computation steps (ACS) [Li et al., 2018]

- Learn how many tokens to generate with encoder outputs

Continuous Integrate-and-Fire (CIF) [Dong et al., 2019]

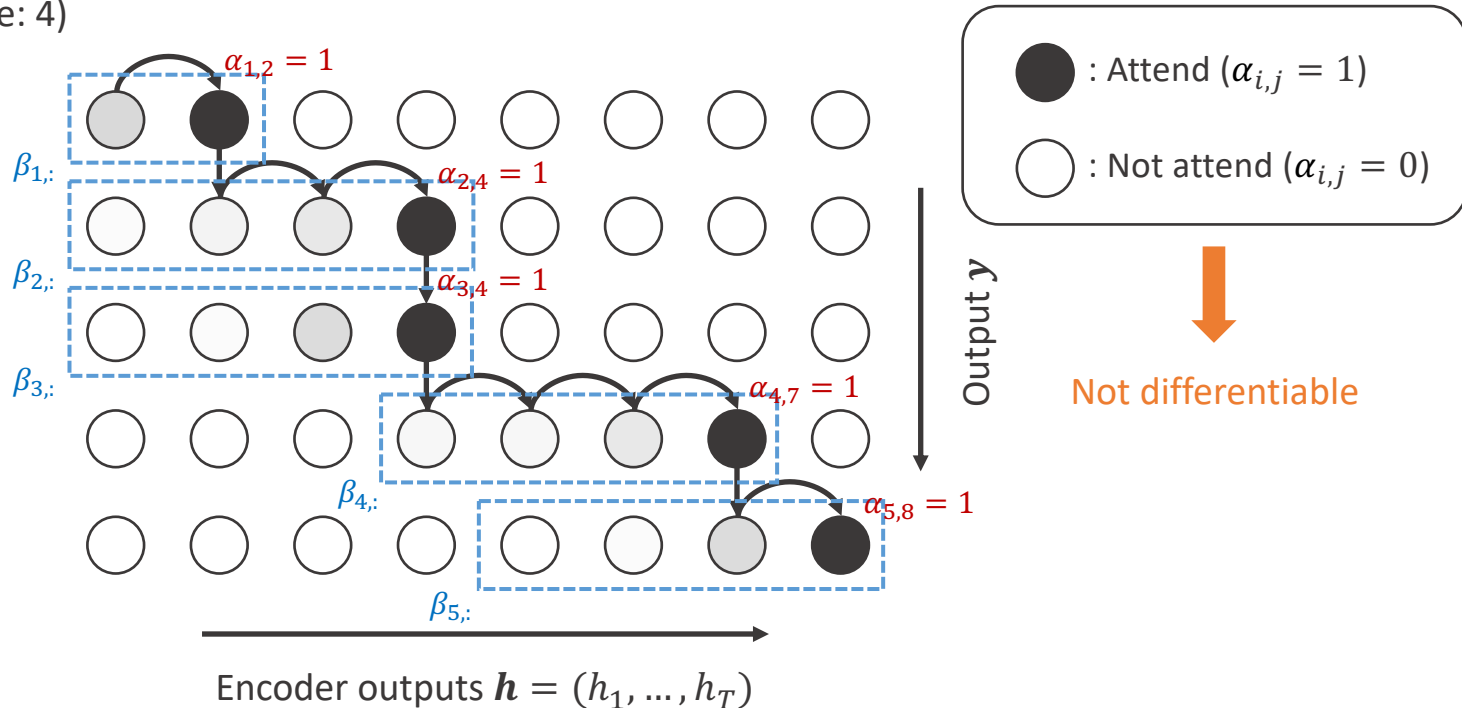
- Fine-grained version of ACS

And more...

- Windowing approaches
- Reinforcement learning

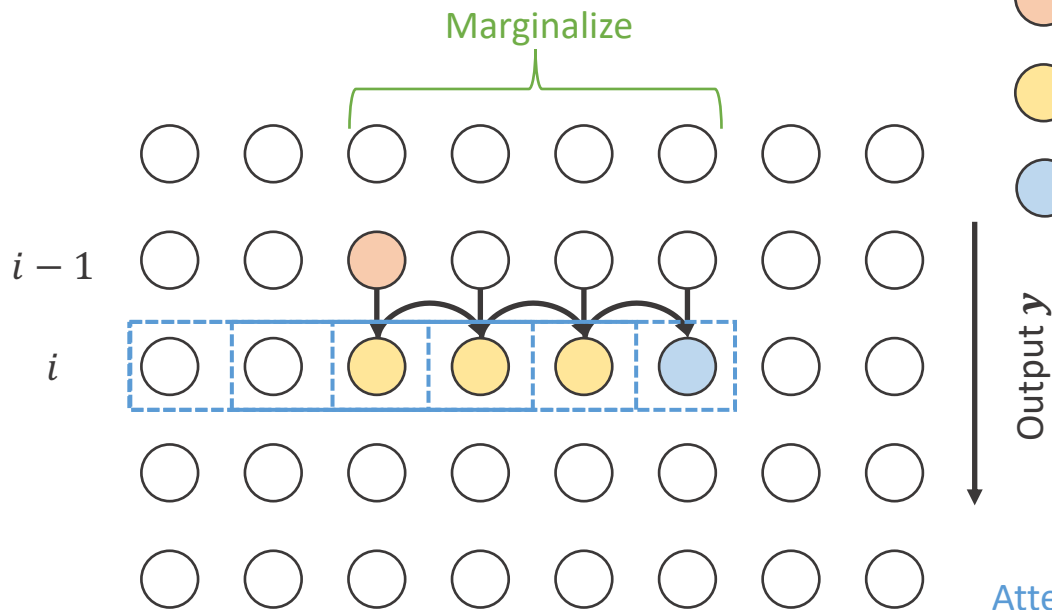
MoChA (test time)

e.g., $w = 4$ (chunk size: 4)



1. **Monotonic attention**: whether to attend or not
2. **Chunkwise attention**: soft attention over a small window

MoChA (training time)



○ : Attend at $(i - 1)$ -th step

○ : Not attend

○ : Attend at i -th step

Use expected alignments
during training
for backpropagation

Encoder outputs $\mathbf{h} = (h_1, \dots, h_T)$

Previous attention

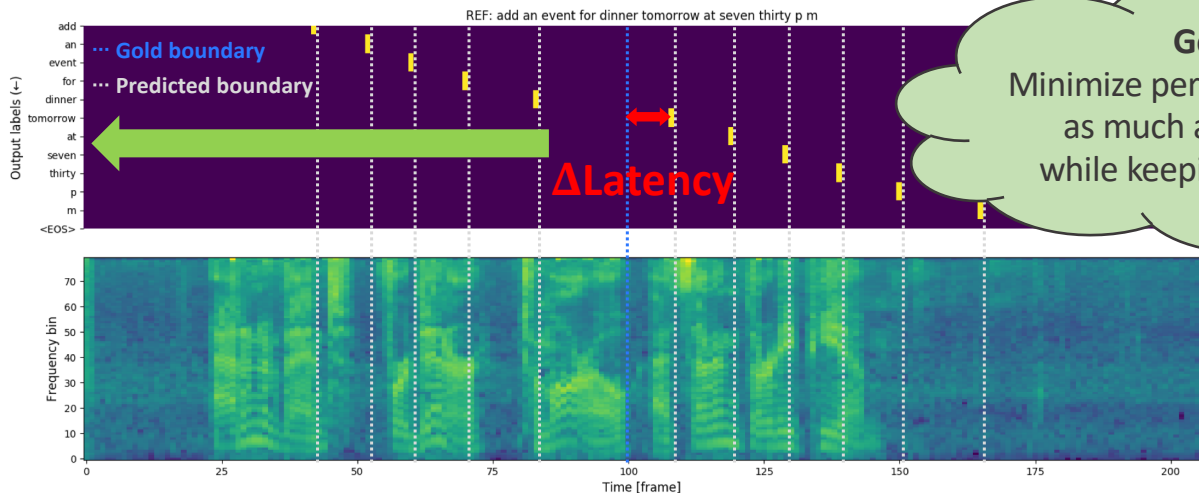
Attend

Not attend

$$\alpha_{i,j} = p_{i,j} \sum_{k=1}^j \left(\alpha_{i-1,k} \prod_{l=k}^{j-1} (1 - p_{i,l}) \right)$$

$$= (1 - p_{i,j-1}) \frac{\alpha_{i,j-1}}{p_{i,j-1}} + \alpha_{i-1,j}$$

Delayed token generation problem



Goal

Minimize perceived latency as much as possible while keeping accuracy

- Decision boundaries (yellow dots) are delayed from the actual acoustic boundary
 1. Unidirectional encoder (lacking the future information)
 2. Sequence-level criterion (utilizing as many future frames as possible to maximize the log-likelihood)
- This leads to increasing user perceived latency
 - Similar behaviors have been reported in CTC [sak et al., 2015] and RNN-T [Li et al., 2019]

Evaluation metric: latency

- Definition: difference between time-index of a predicted boundary and that of the gold boundary

Corpus-level latency (averaged per token)

$$\Delta_{\text{corpus}} = \frac{1}{\sum_{k=1}^N |\mathbf{y}^k|} \sum_{k=1}^N \sum_{i=1}^{|\mathbf{y}^k|} (\widehat{b}_i^k - b_i^k)$$

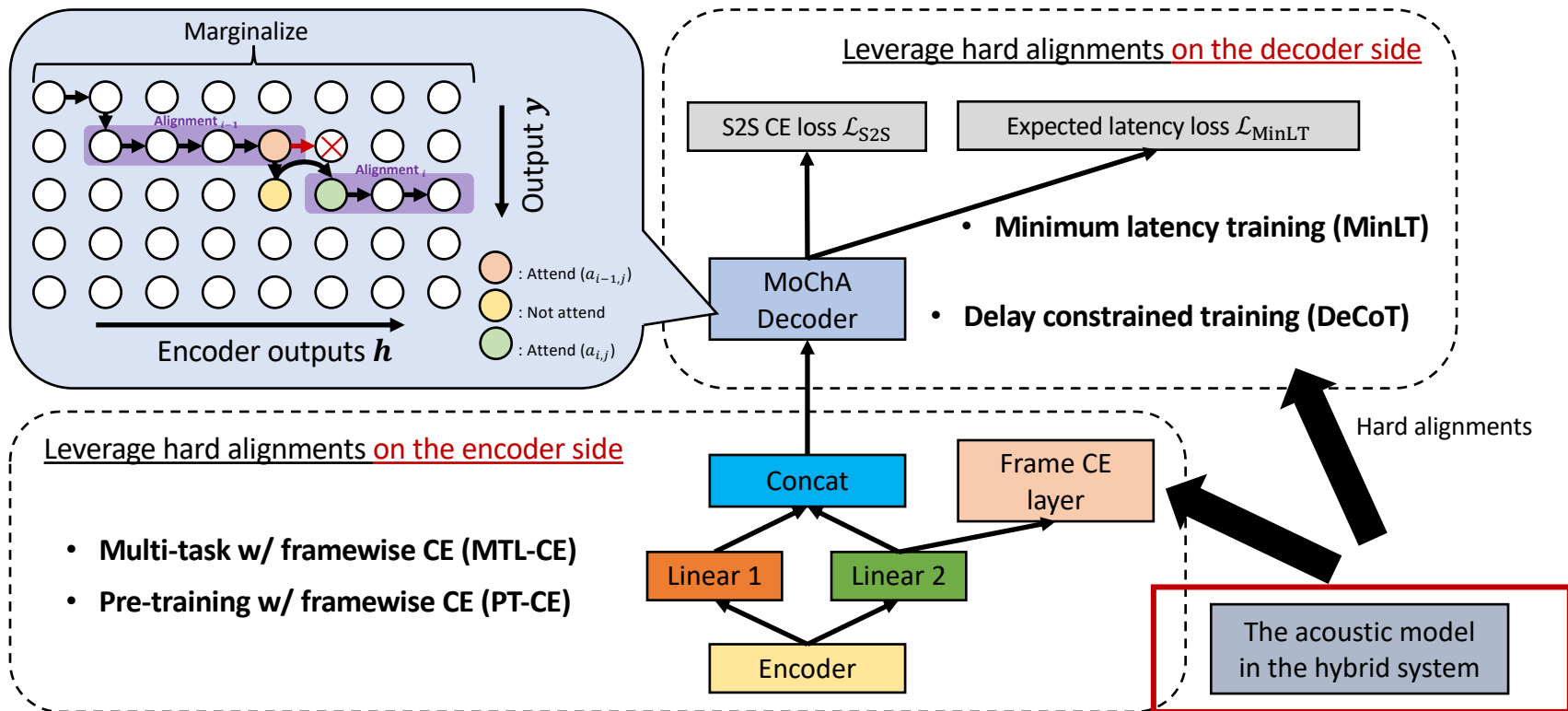
Utterance-level latency (averaged per utterance)

$$\Delta_{\text{utterance}} = \frac{1}{N} \sum_{k=1}^N \frac{1}{|\mathbf{y}^k|} \sum_{i=1}^{|\mathbf{y}^k|} (\widehat{b}_i^k - b_i^k)$$

- Report (1) average, (2) median, (3) 90-th, and (4) 99-th percentile
- Teacher-forcing when calculating latency to match the sequence lengths

Proposed methods

Where should we apply alignment information in the model?



Leveraging hard alignments extracted from the hybrid system

1. Multi-task learning w/ framewise CE (MTL-CE)

◆ Objective function

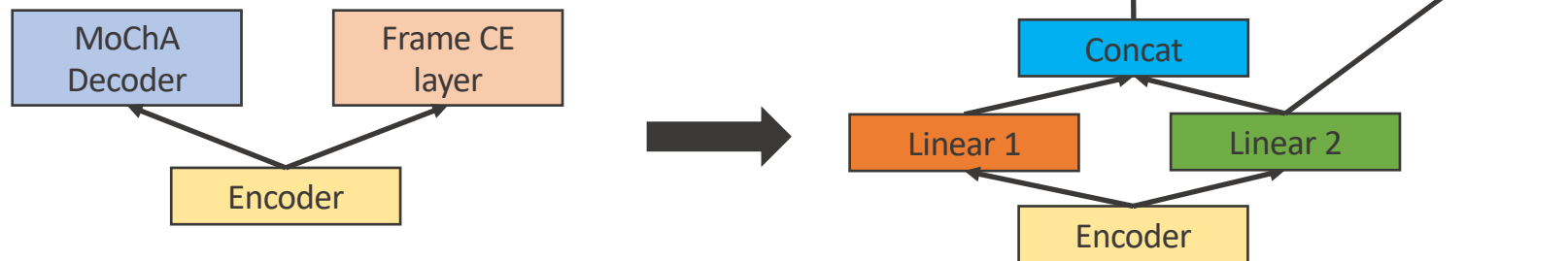
Train both branches
from scratch

$$\mathcal{L}_{\text{total}} = (1 - \lambda_{\text{CE}}) \underbrace{\mathcal{L}_{\text{S2S}}(\mathbf{y}|\mathbf{x})}_{\text{MoChA}} + \lambda_{\text{CE}} \underbrace{\mathcal{L}_{\text{CE}}(\mathbf{A}|\mathbf{x})}_{\text{Frame CE}} \quad (0 \leq \lambda_{\text{CE}} \leq 1)$$

- Motivation: align encoder outputs to the true acoustic location

◆ Insert linear bottleneck layers

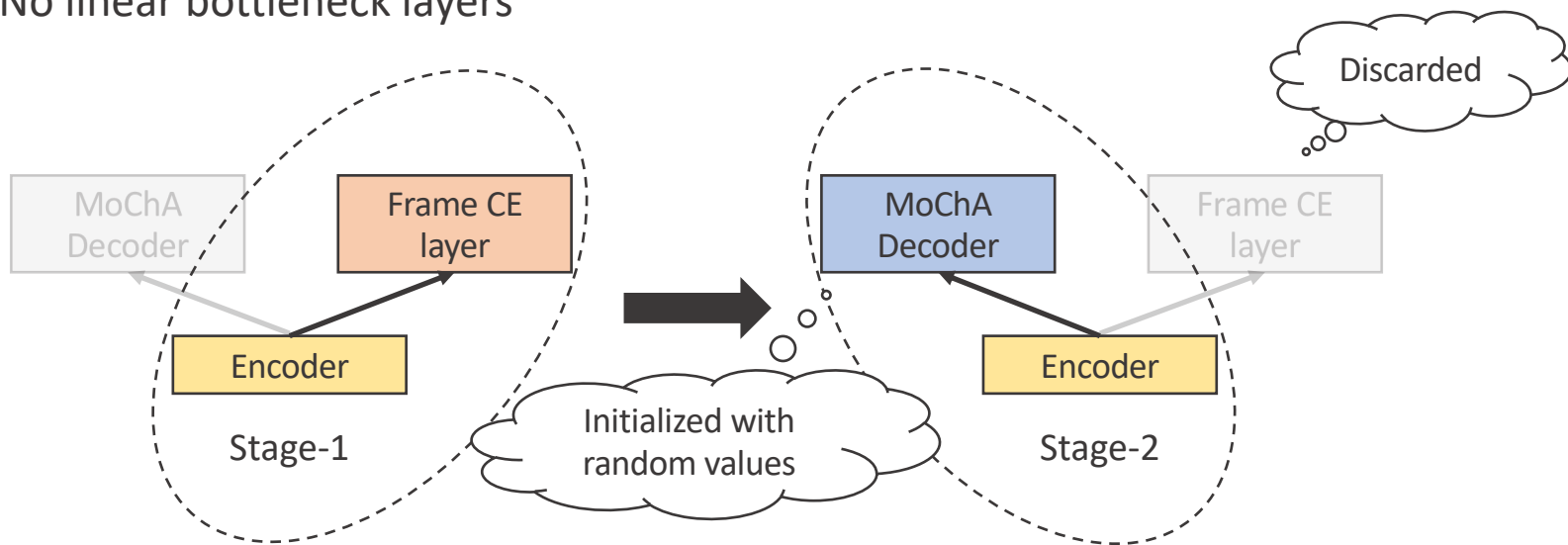
- Inspired by the CTC acoustic model [Yu et al., 2018]



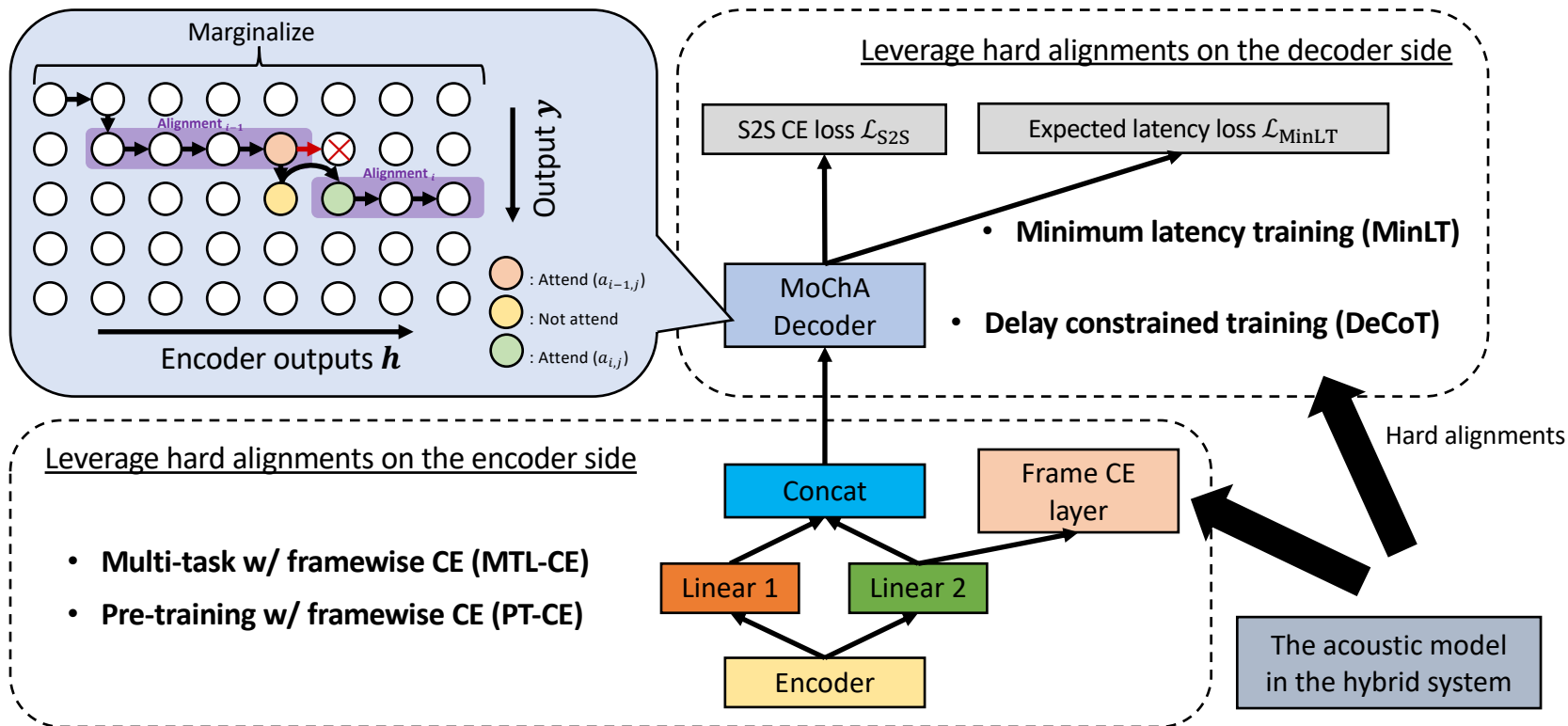
2. Pre-training with framewise CE (PT-CE)

◆ 2-staged training

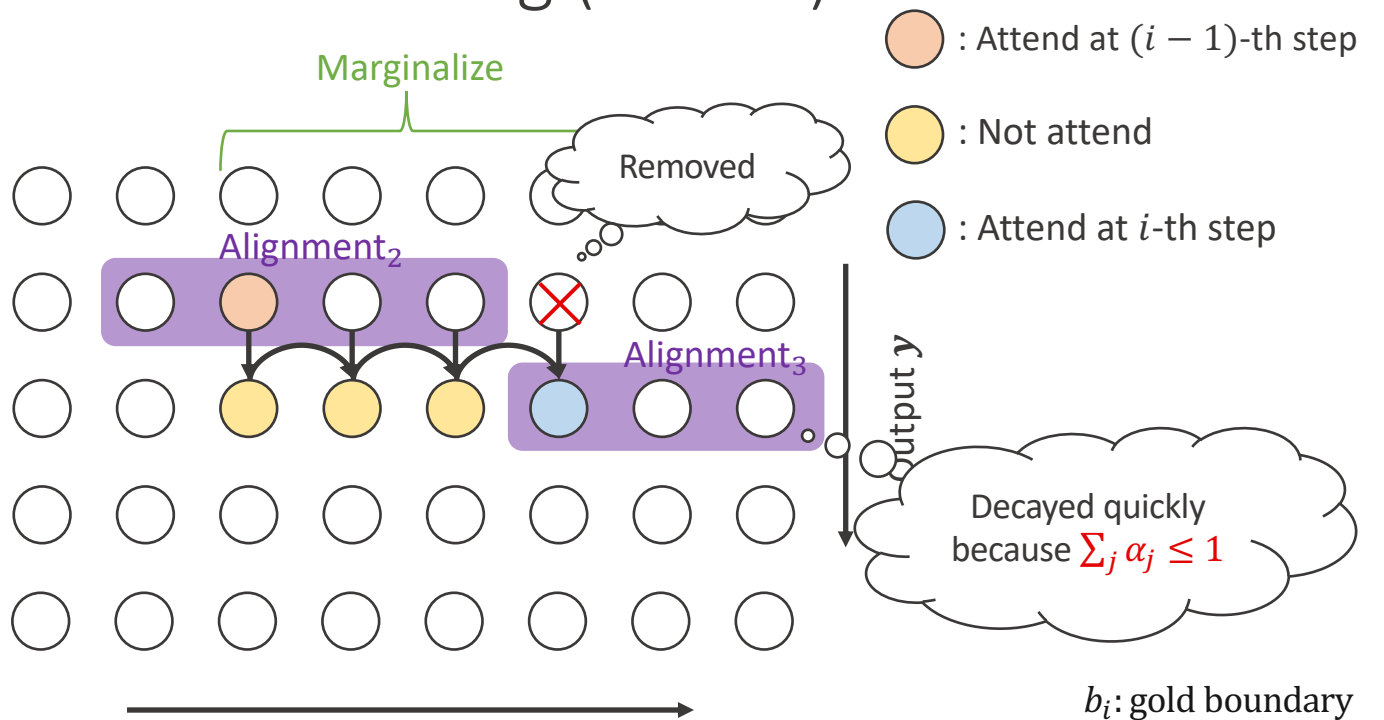
- Motivation
 - Start training from well-aligned encoder representations
 - Do not have to tune the CE weight λ_{CE}
- No linear bottleneck layers



Proposed methods



3. Delay constrained training (DeCoT)



Encoder outputs $\mathbf{h} = (h_1, \dots, h_T)$

Remove inappropriate paths whose boundaries surpass the actual acoustic boundary more than fixed acceptable latency δ [frames]

$$\alpha_{i,j} = \begin{cases} p_{i,j} \left((1 - p_{i,j-1}) \frac{\alpha_{i,j-1}}{p_{i,j-1}} + \alpha_{i-1,j} \right) & (j \leq b_i + \delta) \\ 0 & (\text{otherwise}) \end{cases}$$

3. Delay constrained training (DeCoT)

Regularization with quantity loss

- Add a regularization term to keep $\sum_j \alpha_j = 1$
- Originally proposed in CIF [Dong et al., 2019] with a different motivation

L : the number of tokens in the reference

$$\mathcal{L}_{\text{QUA}} = |L - \sum_{i=1}^L \sum_{j=1}^T \alpha_{i,j}|$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{S2S}} + \lambda_{\text{QUA}} \mathcal{L}_{\text{QUA}} \quad (\lambda_{\text{QUA}} \geq 0)$$

4. Minimum latency training (MinLT)

◆ Objective function

- Directly minimize the expected latency $\mathcal{L}_{\text{MinLT}}$ by utilizing hard alignments A

Expected boundary

$$\mathcal{L}_{\text{MinLT}} = \frac{1}{|y|} \sum_{i=1}^L \left| \sum_{j=1}^T j \alpha_{i,j} - b_i \right| \quad (b_i: \text{reference boundary for } i\text{-th token})$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{S2S}} + \lambda_{\text{MinLT}} \mathcal{L}_{\text{MinLT}} \quad (\lambda_{\text{MinLT}} \geq 0)$$

- Motivation: reduce latency flexibly
 - DeCoT assumes the fixed latency for each token

◆ Related work

- Latency loss has been investigated in simultaneous NMT [Arivazhagan et al., 2019]
- Non-silence frames are not distributed uniformly over the input speech in ASR

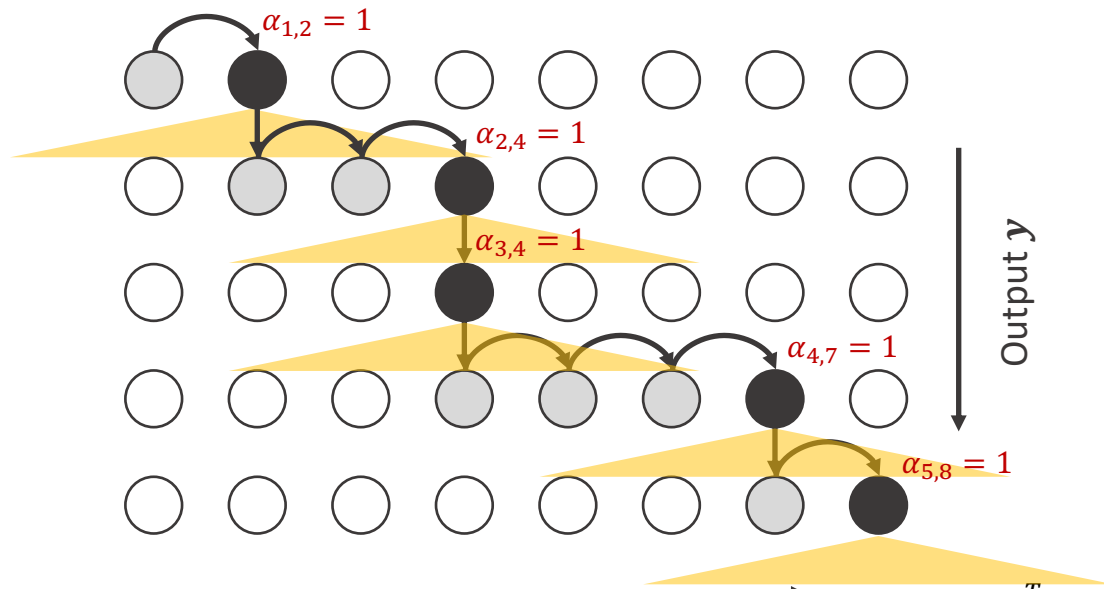
Experimental condition

Data	Train: Cortana voice assistant (3.4k hours) Validation: Sampled disjoint 4k utterances form the training set Test: 5.6k utterances
Feature	80-dim log-mel fbank (3 frame stacked, 30ms per frame)
Output unit	Mixed units (34k vocabulary)
Architecture	Offline: 512-dim (per direction) 6-layer BiGRU encoder Streaming: 1024-dim 6-layer GRU encoder Decoder: 512-dim 2-layer GRU
Optimization	Adam
Decoding	Beam width: 8, no LM

- Word-level alignments: $A = (a_1, \dots, a_T)$ ($\{a_j\}_{j=1, \dots, T}$: one-hot vector)
 - Divide duration based on the ratio of the character length of each subword
- **Start DeCoT and MinLT from the baseline MoChA (warm start training)**

Enhance monotonic attention with 1D convolution

e.g., $k = 5$ (lookahead: 2, 60ms)



Encoder outputs $\mathbf{h} = (h_1, \dots, h_T)$

$$e_{i,j}^{\text{mono}} = g \frac{v^T}{\|v\|} \text{ReLU}(W_h \mathbf{h}'_j + W_s s_i + b) + r$$

$$\mathbf{h}'_j = W_c * h_j \quad (W_c \in \mathbb{R}^{d \times d \times k})$$

k : kernel size, d : unit size

◆ Motivation

- Leveraging the surrounding frames for robust binary decision

Results: Baseline

Model		WER [%]
Offline	BiGRU global attention	7.01
	UniGRU global attention	8.44
	BiGRU MoChA (chunk: 4)	8.09
Streaming	UniGRU MoChA (chunk:4)	10.37
	+ 1D-convolution (baseline)	9.93



4.24% WERR

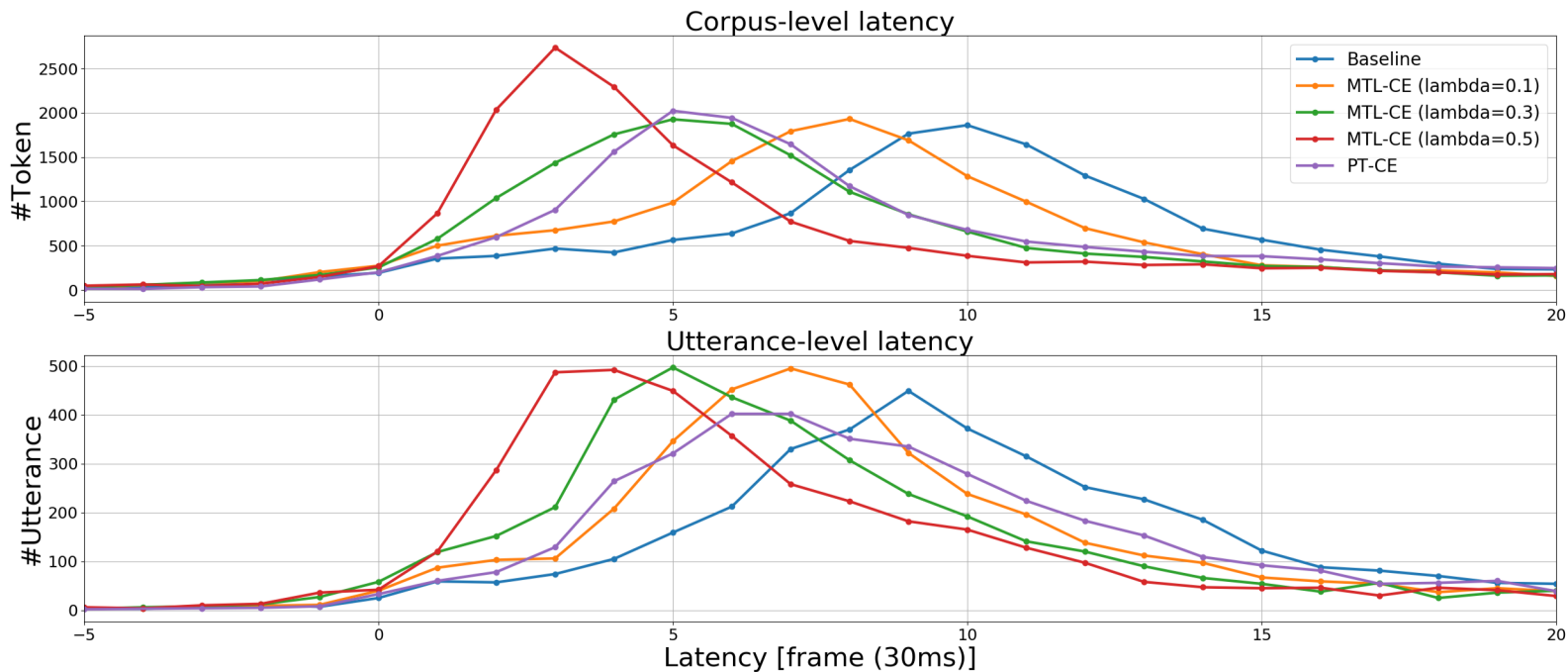
- Huge gaps between (1) bidirectional $\leftrightarrow$ unidirectional
(2) offline $\leftrightarrow$ streaming S2S
- 1D-convolution layer improved the streaming MoChA by **4.24%** relatively

Results: Alignments on the encoder side

Model	WER [%]	Corpus-level [frame (30ms)]			
		Ave.	Med.	90th	99th
Baseline MoChA	9.93	11.65	10.00	21.39	44.29
MTL-CE ($\lambda_{CE} = 0.1$)	10.21 5.6%	9.84	8.00 40%	19.42	46.54
MTL-CE ($\lambda_{CE} = 0.3$)	10.48	8.78	6.00	19.69	47.96
MTL-CE ($\lambda_{CE} = 0.5$)	11.11	8.36	5.00	21.21	49.86
PT-CE	12.74	10.49	7.00	22.90	48.65

- MTL-CE reduced latency in proportion to λ_{CE} while degrading WER slightly
- PT-CE also reduced latency but degraded WER too much
- Contrastive results to previous works using CTC + framewise CE objective
 - MoChA is a label-synchronous model

Visualization of latency distribution (encoder)

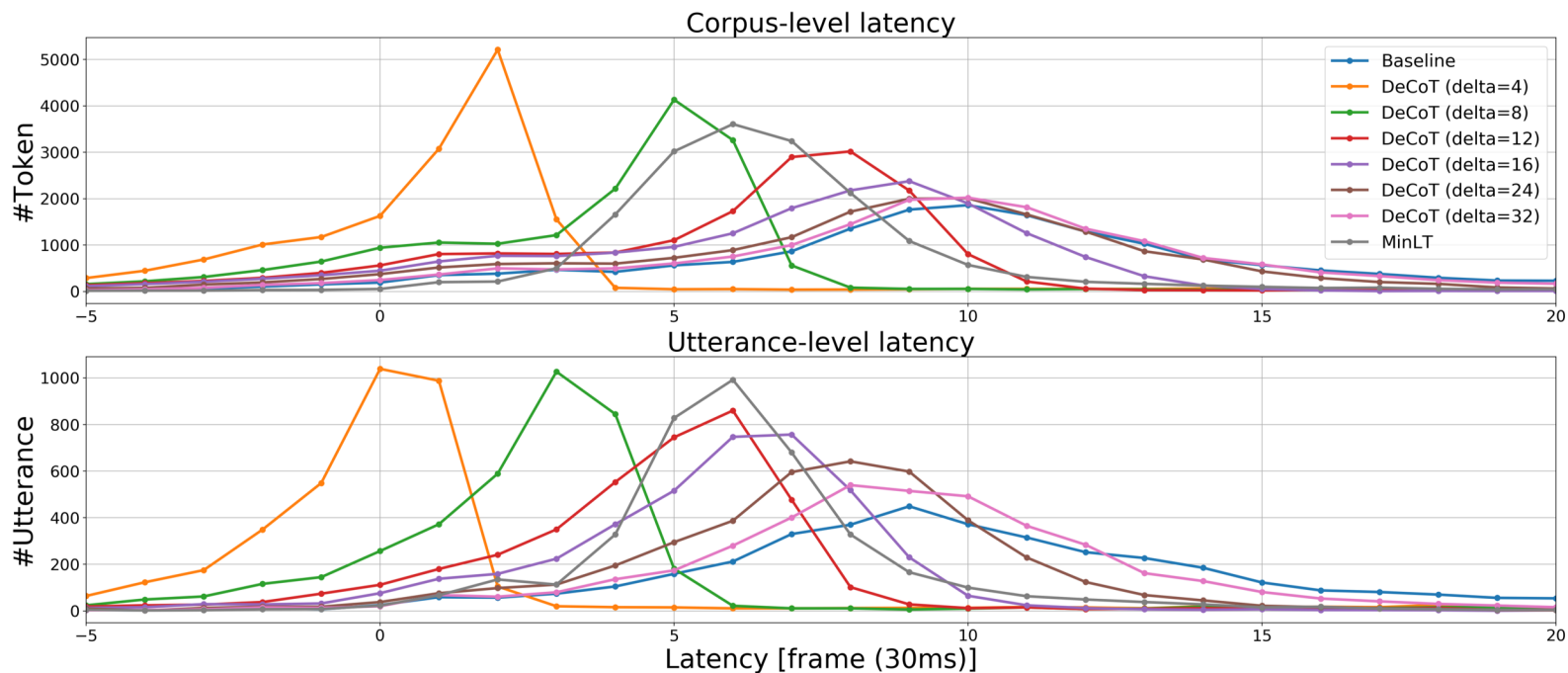


Results: Alignments on the decoder side

Model	WER [%]	Corpus-level [frame (30ms)]			
		Ave.	Med.	90th	99th
UniGRU global attention (offline)	8.44	N/A	N/A	N/A	N/A
Baseline MoChA	9.93	11.65	10.00	21.39	44.29
DeCoT ($\delta = 4$)	20.25	3.66	1.00	9.56	62.27
DeCoT ($\delta = 8$)	14.35	4.60	5.00	7.00	47.04
DeCoT ($\delta = 12$)	11.40	6.02	7.00	9.92	35.58
DeCoT ($\delta = 16$)	9.13	6.63	8.00	11.71	16.43
DeCoT ($\delta = 24$)	8.87	8.37	9.00	14.45	21.07
DeCoT ($\delta = 32$)	9.17	9.79	10.00	16.54	27.01
MinLT	9.70	7.06	6.00	10.63	26.76

- DeCoT: large WER improvement and moderate latency reduction (tail part)
- MinLT: small WER improvement and large latency reduction (median)

Visualization of latency distribution (decoder)



Ablation study: Decoder side

- **Combination of DeCoT and MinLT reduced the latency, but degraded WER too much**

Model	WER [%]	Corpus-level [frame (30ms)]			
		Ave.	Med.	90th	99th
DeCoT ($\delta = 16$)	9.13	6.63	8.00	11.71	16.43
+ MinLT	12.75	4.05	4.00	7.96	15.92
MinLT	9.70	7.06	6.00	10.63	26.76

Ablation study: Decoder side

- **Quantity loss was essential for DeCoT but not necessary for the baseline and MinLT**

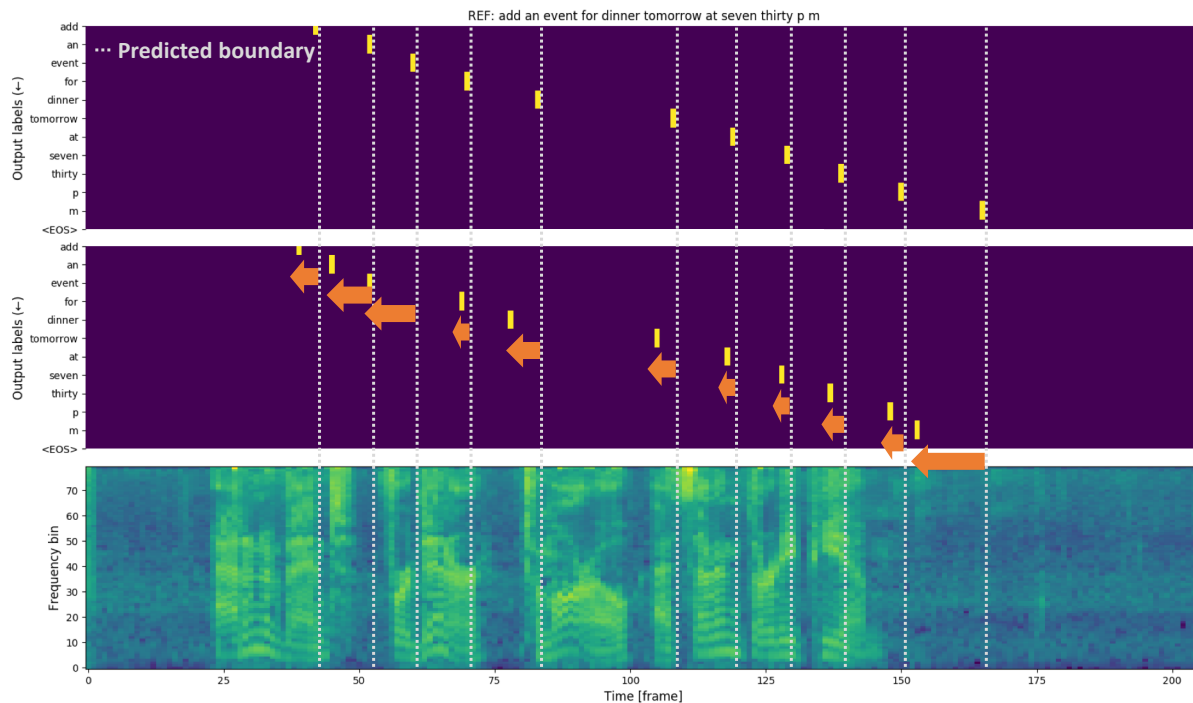
Model	WER [%]	Corpus-level [frame (30ms)]			
		Ave.	Med.	90th	99th
Baseline MoChA	9.93	11.65	10.00	21.39	44.29
+ Quantity loss	10.30	11.24	10.00	20.39	36.01
DeCoT ($\delta = 16$)	9.13	6.63	8.00	11.71	16.43
- Quantity loss	14.28	3.93	3.00	7.20	27.39
MinLT	9.70	7.06	6.00	10.63	26.76
+ Quantity loss	13.66	6.82	6.00	10.45	25.57

Ablation study: Decoder side

- **Warm start training from the baseline was effective for DeCoT and MinLT**

Model	WER [%]	Corpus-level [frame (30ms)]			
		Ave.	Med.	90th	99th
Baseline MoChA	9.93	11.65	10.00	21.39	44.29
+ Warm start training	9.21	12.27	11.00	22.23	43.16
DeCoT ($\delta = 16$)	9.13	6.63	8.00	11.71	16.43
- Warm start training	10.72	6.28	7.00	11.12	36.03
MinLT	9.70	7.06	6.00	10.63	26.76
- Warm start training	13.63	11.83	10.00	21.41	45.06

Alignment visualization

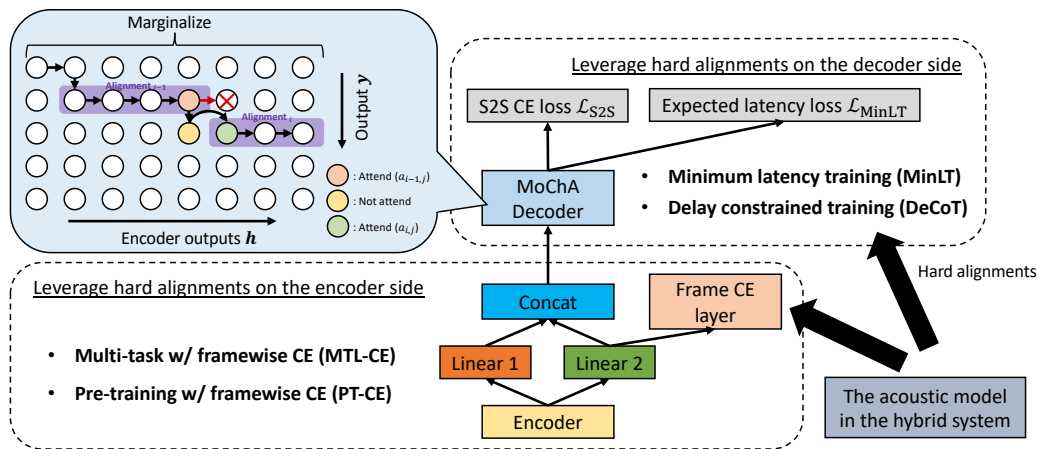


Baseline MoChA

DeCoT ($\delta = 16$)

Conclusion

- Explored to leverage frame-level hard alignments extracted from the hybrid system to reduce user perceived latency
- Alignments were effective for latency reduction on both sides, and also improved ASR performance when applying on the decoder side



Question?

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